

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

*I _C.C. Jyoshna (192411098)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C.C.Jyoshna

Q26. Compare the performance of 3-DES against AES-128 for encrypting a 10 GB database backup, and explain which algorithm an architect should choose for a new system in 2026.

Introduction:

3-DES and AES-128 are symmetric encryption algorithms used to protect sensitive data. However, they differ significantly in performance, security, and efficiency.

Comparison

3-DES	AES-128
Encrypts data three times	Encrypts data once
Slower execution	Faster execution
64-bit block size	128-bit block size
Higher power consumption	Lower power consumption
Older standard	Modern encryption standard

Example:

Suppose a company needs to encrypt a **10 GB database backup**.

- Using **3-DES**, the backup takes longer because every data block is encrypted three times.
- Using **AES-128**, the backup completes much faster while providing strong security.

Recommendation:

For a **new system in 2026**, **AES-128** should be selected because it is faster, more secure, energy-efficient, and recommended by modern security standards.

Conclusion:

AES-128 is the preferred algorithm for new applications because it offers better performance and stronger security than 3-DES.

Q27. A company's compliance officer asks why 3-DES, despite being more secure than DES, is still being phased out. Provide a technical explanation citing both performance and block-size vulnerabilities.

Introduction:

3-DES was introduced to improve the security of DES by applying the DES algorithm three times. Although it is stronger than DES, it is no longer recommended for modern applications.

Reasons for Phasing Out 3-DES:

- Encrypting data three times makes 3-DES much slower than modern algorithms such as AES.
- It uses a **64-bit block size**, which increases the possibility of block collisions when large amounts of data are encrypted.
- Modern security standards recommend AES because it provides better security and performance.

Example:

Suppose a banking server encrypts millions of transactions every day.

- With **3-DES**, encryption is slower and repeated blocks may eventually create security risks.
- With **AES**, encryption is faster and the larger block size provides stronger protection.

Conclusion:

Although 3-DES is more secure than DES, it is being replaced because AES provides higher performance and better protection against modern security threats.

Q28. An attacker intercepts a large volume of 3-DES-CBC encrypted traffic between two servers exceeding 4 GB. Explain the practical risk this poses due to block collisions.

Introduction:

3-DES uses a **64-bit block size**. When very large amounts of data are encrypted using the same key, repeated ciphertext blocks may occur due to block collisions.

Practical Risk:

- Encrypting more than approximately **4 GB** of data increases the chance of repeated ciphertext blocks.
- An attacker observing these repeated blocks may identify patterns in the encrypted communication.
- Although the attacker cannot immediately decrypt the data, repeated blocks may help in cryptanalysis and reduce overall security.

Example:

Suppose two servers exchange **6 GB** of encrypted backup data using the same 3-DES key.

Since the amount of encrypted data exceeds the safe limit, repeated ciphertext blocks may appear, increasing the possibility of block collisions.

Recommendation:

- Use **AES-128 or AES-256** with a **128-bit block size**.
- Change encryption keys regularly to reduce collision risk.

Conclusion:

Large data transfers using 3-DES increase the risk of block collisions, making AES a safer choice for modern systems.

Q29. A legacy hardware security module (HSM) only supports 3-DES for key wrapping operations. Propose a mitigation strategy while a full migration to AES-based key wrapping is planned.

Introduction:

Some legacy HSMs still support only 3-DES for key wrapping. Replacing such systems immediately may not be practical.

Mitigation Strategy:

Continue using 3-DES only for key wrapping until migration is completed.

Limit the amount of data encrypted with the same key.

Rotate encryption keys regularly.

Protect communication using secure protocols such as TLS.

Plan a phased migration to AES-based key wrapping.

Example:

Suppose a bank's HSM currently supports only 3-DES.

Instead of replacing the entire system immediately, the bank continues using 3-DES for key wrapping while gradually upgrading all systems to AES during scheduled maintenance.

Conclusion:

A phased migration minimizes operational risk while improving long-term security through AES adoption.

Q30. Explain to a junior developer why 3-DES applies DES three times instead of simply increasing DES to a single pass with a 168-bit key.

Introduction:

DES was originally designed as a **56-bit encryption algorithm**. Its internal structure cannot simply be converted into a 168-bit version by increasing the key size.

Explanation:

- The DES algorithm was designed to work with a 56-bit key.
- Simply increasing the key size would not significantly improve its security because the internal design remains the same.
- 3-DES improves security by applying DES three times with different keys (Encrypt–Decrypt–Encrypt).
- This approach strengthens encryption while remaining compatible with existing DES hardware and software.

Example:

Imagine a door protected by **one lock**.

Using a larger key for the same lock does not change how the lock works.

Adding **three separate locks** provides much stronger protection because an intruder must unlock all three.

Similarly, 3-DES applies DES three times to increase security.

Conclusion:

3-DES improves security by performing three DES operations instead of modifying the original DES algorithm, making it more secure while maintaining compatibility with existing systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand